

Mihika Sharma

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EDUCATION

Purdue University

Bachelor of Science in Computer Science, Cert. in Applications of Data Science

West Lafayette, IN

Aug. 2022 – May 2026

EXPERIENCE

Software Engineer Intern

Aug. 2025 – May 2026

Union Pacific Railroad

Hybrid

- Built a Python and Tableau tabcmd backup pipeline to automate daily executive report delivery and reduce dependency on a failure-prone UiPath workflow.
- Developed a Tableau write-back extension in Python Flask and JavaScript to migrate legacy ColdFusion operational workflows directly into Tableau dashboards used by internal teams.
- Implemented role-based permissions, session management, auto-login authentication, audit logging, configurable SQL actions, and automatic dashboard refresh after write operations.

Software Engineer Intern

May 2025 – Aug. 2025

Union Pacific Railroad

Omaha, NE

- Built an ETL pipeline to ingest and geolocate-match 88K+ external facilities against internal Teradata using fuzzy spatial logic via external APIs.
- Built a caching system in SQL Server + T-SQL to avoid redundant processing across data exports, reducing run time by 48% and external API calls by 60%.
- Uncovered 30+ new leads worth \$750k in projected revenue, and delivered results to stakeholders via ArcGIS + Tableau dashboards.

Software Developer

Aug. 2024 – Dec 2025

GAINS AI

West Lafayette, IN

- Built an Unreal Engine simulation tool that enables urban planners to test pedestrian flow in public space designs.
- Used Environmental Query Systems and Behavior Trees to simulate crowds of 200+ AI-driven NPCs with dynamic traits like cone of vision, sun preference, and crowd affinity.
- Wrote C++ scripts to log and export 10K+ NPC movement data points for GIS-based heatmap analysis.

Software Engineer Intern

July 2024 – Aug. 2024

Saarathi.ai

Delhi, India

- Led the redesign of a digital lending platform to reduce user journey completion time from 30+ min to ~10 min.
- Implemented 5 RESTful APIs in Java Spring Boot to enable partner data sync and seamless customer onboarding.
- Collaborated across product and engineering teams to design new user flow wireframes in Figma and conducted time-efficiency analysis to secure stakeholder approval for production deployment.

PROJECTS

MazeFindVR | Unity, C#, VR Development, Oculus SDK

May 2025 – Dec. 2025

- VR spatial-navigation system with procedural maze generation, adaptive difficulty scaling, timed gameplay, and performance logging.

InQueue | React, Flask, Firebase, Firestore, Google Maps API

May 2025 – Dec. 2025

- Full-stack virtual queue management platform with real-time queue tracking, location-based queue discovery and business analytics for high-wait environments.

SFSL Inventory | MongoDB, Express.js, React, Node.js

Aug. 2024 – May 2025

- Full-stack inventory management dashboard for a non-profit food bank with Prophet-based demand forecasting.

BoilerTrack | SReact, Flask, SQLite, Python, JavaScript, Git

Aug. 2024 – Dec. 2024

- Centralized lost and found system featuring item matching, QR code-based claims, and real-time tracking.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Swift, SQL, JavaScript, HTML/CSS, R

Frameworks: React, Flask, Node.js, Express.js, Spring Boot, SwiftUI, Firebase

Developer Tools: Git, Tableau, Google Cloud, SSIS, Postman, Docker, VS Code, Xcode, Maven, Perforce

Libraries: Firestore, Pandas, NumPy, Prophet, PyTorch, REST APIs, Google Maps API, Oculus SDK